

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

LABORATORY ASSIGNMENT REPORT ASSIGNMENT 3: CUSTOM KEYWORDS AND LIBRARIES

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Department	:	CSE (AI & ML)
Course	:	Python for Automation
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1. MODULE NAME & EXPERIMENT TITLE

Module Name: Custom Keywords and Libraries

Experiment Title: Creation of Custom Python Keywords and String/Mathematical Operations Using standard BuiltIn Libraries in Robot Framework

2. PROBLEM STATEMENT

Develop test scripts and custom extensions in Robot Framework to fulfill the following requirements:

- Build a custom Robot Framework keyword using Python to perform a specific action, such as calculating the sum of two numbers.
- Utilize the standard "BuiltIn" library to manipulate strings or perform mathematical operations within test cases.

3. OBJECTIVE

- To extend Robot Framework functionality by writing custom keyword functions in external Python modules.
- To import custom Python classes/libraries directly into Robot Framework test suites under the `*** Settings` header.
- To master built-in utility keywords provided by Robot Framework's default BuiltIn library (e.g., Evaluate, Convert To Lowercase, Should Be Equal As Numbers, Catenate).

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language:** Python 3.11.x
- **Automation Framework:** Robot Framework 6.1
- **Libraries Used:** BuiltIn (Standard Library), CustomMathLibrary.py (Custom Python Library)
- **IDE / Development Environment:** Visual Studio Code (VS Code)
- **Core Concepts:** Python Method-to-Keyword mapping, Dynamic Typing Conversion, Expression Evaluation, String Manipulation Functions.

5. SOURCE CODE / IMPLEMENTATION STEPS

Step 1: Custom Python Library File (CustomMathLibrary.py)

CustomMathLibrary.py - Python Custom Keyword Source

Python

```
1 # Custom Python Library for Robot Framework

2 class CustomMathLibrary:

3     def calculate_sum(self, num1, num2):

4         """Calculates the sum of two numbers and returns the result."""

5         result = float(num1) + float(num2)

6         return result
```

6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The code was executed in VS Code terminal using the command `robot assignment3_tests.robot`. Robot Framework automatically parsed the custom Python class methods as test keywords and executed all string/math assertions.

Screenshot 1: VS Code Integrated Terminal Execution Output

```
PS C:\Users\Ashmita\Automation_Lab> robot assignment3_tests.robot
===== Assignment3 Tests
===== Calculated Sum:
40.0 TC_01: Execute Custom Python Keyword For Addition | PASS | -----
----- Processed Name: ASHMITA SAHA TC_02: String
Manipulation Using BuiltIn Library | PASS | -----
----- Expression Output: 36.5 TC_03: Mathematical Operations Using BuiltIn
Evaluate | PASS | -----
Assignment3 Tests | PASS | 3 tests, 3 passed, 0 failed
===== Output:
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html Report:
C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

Screenshot 2: Generated HTML Execution Summary (report.html)

Robot Framework Test Suite Status: ALL 3 TESTS PASSED

Suite Name:	Assignment3 Tests
Loaded Libraries:	CustomMathLibrary.py, BuiltIn
Execution Time:	00:00:00.285 seconds
Verified Operations:	Custom Python keyword addition ('Calculate Sum'), String Catenate & Upper, Python expression evaluation ('Evaluate').

7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

Brief Result:

All 3 test cases executed without errors. The custom keyword `Calculate Sum` defined in Python correctly added two floating-point numbers, and the standard `BuiltIn` keywords successfully manipulated string strings and mathematical expressions.

Observation:

- Robot Framework automatically converts Python method names from `snake\_case` (e.g., `calculate\_sum`) into title case keywords (e.g., `Calculate Sum`).
- Importing custom Python classes requires providing the exact relative file path (`CustomMathLibrary.py`) under the `\*\*\* Settings \*\*\*` section.
- The `Evaluate` keyword from the `BuiltIn` library allows executing raw Python expressions directly inside `.robot` files without needing separate custom files for minor arithmetic logic.

Conclusion:

In this lab assignment, I learned how to create custom Python libraries to extend Robot Framework's capability beyond built-in web browser automation keywords. Additionally, I explored string manipulation and math functions using the standard `BuiltIn` library. This modular approach allows writing clean, readable, and highly extensible automated testing workflows.